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Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China

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Abstract

We examine how the spatial organisation of humid heat changes when regional wet-bulb temperature is high. Synchronous fields over eastern China are compared within summer months across 33 evaluation summers. A multiscale graph analysis expresses regional dispersion and its spatial components within the same relative contrast. The mean squared pairwise contrast is 7.28% lower on high-mean days, with attenuation concentrated in the broad latitudinal structure. Broad-scale anomaly energy changes little; the contraction is associated with anomalies opposing the monthly climatological gradient. On a nested observation grid, these days also exhibit a wider connected humid-heat footprint. At 24°C, total and largest connected exceedance fractions are each approximately 13 percentage points greater than on middle days. The coverage difference varies across temperature thresholds. Simple spatial summaries retain most coverage information, while the graph profile identifies the scales and structures of the dispersion change. Historical and station comparisons support the broad pattern. The results distinguish attenuation of persistent geographic contrasts from a reduction in transient anomaly variability, establishing a structural interpretation of regional humid-heat coherence.

Keywords ERA5-Land, graph Laplacian, humid heat, randomisation inference, semivariance, spatial statistics

1. Introduction

Wet-bulb temperature (WBT) is a widely used measure of humid heat stress that combines air temperature and humidity (Sherwood and Huber, 2010; Raymond et al., 2020). Two days can have identical regional means yet sharply different spatial fields: one may contain strong north–south or local contrasts, while the other is comparatively coherent.

Recent work has documented increasingly concentrated humid-heat extremes in standard climate regions (Speizer et al., 2022), sharp local wet-bulb extremes associated with wet soils (Chagnaud et al., 2025), and reduced intra-urban heat-stress variability during some heat waves (Shreevastava et al., 2023). Spatial-extremes models such as Healy et al. (2025) estimate nonstationary temperature tails and the area exceeding critical thresholds. The within-month comparison here examines how pairwise WBT differences change with graph scale. Ordinary cross-site variance combines neighbouring differences, mesoscale structure and the domain-wide gradient in a single value. Empirical variograms represent distance-dependent variation, and kernel smoothing has been used to estimate isotropic semivariograms nonparametrically (García-Soidán et al., 2004). Here, Gaussian weights define a multiscale regime-contrast summary for observed fields across the prespecified distance scales. Smooth weighting provides this summary and permits an exact spatial allocation of the regional contrast.

We ask how the spatial pattern of wet-bulb temperature changes on days with high regional means. In eastern China, humid heat varies along a persistent north–south gradient as well as over shorter distances. A reduction in the regional variance could arise from a weaker geographic gradient, smaller local fluctuations, or both. Distinguishing these changes establishes how regional humid-heat conditions relate to the distribution of spatial contrasts. Topography (Pepin et al., 2015), soil moisture (Seneviratne et al., 2010) and circulation (Horton et al., 2015) provide physical context for these patterns.

The area above a temperature level and the size of its largest connected region describe how extensively humid heat occurs at one time. We measure both quantities across a fixed temperature range and examine their relationship to the spatial contrasts. We also assess how much information about these dense-grid states is retained by summaries from the original 121 sites, comparing graph scales with variance, latitude gradient and binned semivariances.

We formulate the comparison using Gaussian-weighted graph semivariances, building on the connection between graph Dirichlet energy and spatial variation (Cressie, 1993; Shuman et al., 2013; Dong et al., 2016). The same relative contrast is retained across bandwidths, site allocations and structural projections. This connects an aggregate dispersion change to the geography of the field and separates the climatic gradient from daily anomaly energy. Event labels and bandwidths are fixed before evaluation to separate specification from the contrasts being assessed (Simmons et al., 2011; White, 2000).

The application identifies the weakening of a broad climatic gradient as the principal spatial feature of high regional humid heat. Direct coverage measurements establish the accompanying geographic extent, and comparisons with simple summaries assess the information supplied by spatial scale. The historical record and station observations provide complementary evidence about the persistence of the pattern. The contribution is the resulting statistical interpretation of regional coherence through its spatial components and its observed humid-heat footprint.

2. Multiscale spatial dispersion

2.1. Graph dispersion

Let y_{it} denote WBT at location i in the selected field for day t, and let $\bar{y}_t = n^{-1}\sum_i y_{it}$. The benchmark is the site mean squared spatial deviation,

$$V_t = \frac{1}{n}\sum_{i=1}^{n}(y_{it} - \bar{y}_t)^2, \qquad n = 121. \tag{1}$$

To retain spatial arrangement, we use an equirectangular kilometre projection centred at the mean site latitude: longitude is multiplied by $111.32\cos(\bar{\phi})$ and latitude by 110.57. For bandwidth h, set $w_{ij,h} = \exp\{-d_{ij}^2/(2h^2)\}$ for $i \neq j$ and $w_{ii,h} = 0$. Let $\mathbf{W}_h = (w_{ij,h})$, $\mathbf{L}_h = \mathrm{diag}(\mathbf{W}_h\mathbf{1}) - \mathbf{W}_h$, and $S_h = \sum_{i<j} w_{ij,h}$. We define graph dispersion as

$$Q_h(\mathbf{y}) = \frac{\mathbf{y}^\top \mathbf{L}_h \mathbf{y}}{2S_h} = \frac{\sum_{i<j} w_{ij,h}(y_i - y_j)^2}{2\sum_{i<j} w_{ij,h}}. \tag{2}$$

Graph dispersion is therefore one half of the edge-weighted mean squared pairwise difference, equivalently an edge-weighted semivariance. Dividing by S_h makes the measure comparable across graphs with different total edge weights. Smaller values indicate greater graph-scale coherence. It has units $^\circ\mathrm{C}^2$; $D_h^{\mathrm{RMS}} = (2Q_h)^{1/2}$ is the weighted root mean squared pairwise WBT difference in $^\circ\mathrm{C}$.

Equation (2) connects three familiar descriptions of spatial variation. It is the graph Dirichlet energy used to measure graph signal roughness (Shuman et al., 2013; Dong et al., 2016). If the field is second-order stationary with semivariogram $\gamma(d_{ij}) = \frac{1}{2}\,\mathrm{E}(Y_i - Y_j)^2$, then

$$\mathrm{E}\{Q_h(\mathbf{Y})\} = \frac{\sum_{i<j} w_{ij,h}\gamma(d_{ij})}{S_h}, \tag{3}$$

so its expected profile is a kernel average of the semivariogram. Under second-order stationarity, this identity supplies a semivariogram interpretation. We calculate Q_h directly from each observed WBT field. On a complete graph with equal off-diagonal weights, $Q_h = nV/(n-1)$. Ordinary spatial variance is therefore the complete-graph endpoint of the same family, up to its degrees-of-freedom factor.

We use five bandwidths selected during method development: $h/d_{\mathrm{med}} \in \{0.125, 0.25, 0.5, 1, 2\}$, where d_{med} is the median pairwise distance. Write these physical bandwidths as h_k, $k = 1, \ldots, H$, with $H = 5$; k is a scale index and h_k is a distance in kilometres. They correspond to 126, 252, 503, 1,006 and 2,013 km. Their weighted mean pair distances are 200, 319, 549, 826 and 992 km, and their effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. Thus 126 km denotes a Gaussian bandwidth with no hard distance cut-off; it is local only relative to the sampled support, where the median nearest-neighbour distance is 150 km. For edge weights w_e, the reported Kish quantity is $N_{\mathrm{eff}} = (\sum_e w_e)^2 / \sum_e w_e^2$.

Let $r = (y, m)$ identify a year and calendar month, with $m \in \mathcal{J} = \{6, 7, 8\}$. Define the set of retained days $\mathcal{T}_{ym} = \{t : t \text{ belongs to year } y \text{ and month } m\}$. For $p \in (0, 1)$, let $q_{p,ym}$ be the type-7 sample quantile of the collection $\{\bar{y}_t : t \in \mathcal{T}_{ym}\}$. The three disjoint sets are

$$\begin{aligned} \mathcal{E}_{ym} &= \{t \in \mathcal{T}_{ym} : \bar{y}_t \geq q_{0.75,ym}\}, \\ \mathcal{M}_{ym} &= \{t \in \mathcal{T}_{ym} : q_{0.25,ym} \leq \bar{y}_t < q_{0.75,ym}\}, \\ \mathcal{L}_{ym} &= \{t \in \mathcal{T}_{ym} : \bar{y}_t < q_{0.25,ym}\}. \end{aligned} \tag{4}$$

Only $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ enter the primary comparison; $\mathcal{L}_{ym}$ denotes the excluded low group. A record subscript r always abbreviates (y, m): for example, $R_{rh} = R_{ymh}$ and $c_{irh} = c_{iymh}$. The thresholds follow each record's regional-mean distribution. We examine within-month seasonal progression through calendar matching. There were no threshold ties; each record has eight high days and 14 or 15 middle days. Quartiles define a relative warm state against the middle half of the monthly distribution, providing eight high days and at least 14 days for the reference mean. Sensitivity analyses replace both cut points, including an upper-tertile versus middle-tertile comparison, while keeping the fields, graph scales and aggregation fixed (Supplement, Section S12).

The two groups are compared at each scale by the relative difference of their dispersion means. Let $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ be the high- and middle-day sets defined above, and let $\overline{Q}^{E}_{ymh}$ and $\overline{Q}^{M}_{ymh}$ denote the corresponding graph-dispersion means, for example $\overline{Q}^{E}_{ymh} = |\mathcal{E}_{ym}|^{-1} \sum_{t \in \mathcal{E}_{ym}} Q_h(\mathbf{y}_t)$, with the middle mean defined analogously. The scale-specific relative contrast and its summer average are

$$R_{ymh} = \frac{\overline{Q}^{E}_{ymh}}{\overline{Q}^{M}_{ymh}} - 1, \qquad \widehat{\theta}_y = \frac{1}{|\mathcal{J}| H} \sum_{m \in \mathcal{J}} \sum_{k=1}^{H} R_{ymh_k}. \tag{5}$$

For a set $\mathcal{Y}$ of summers, the finite-record target is

$$\widehat{\theta}_{\mathcal{Y}} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \widehat{\theta}_y. \tag{6}$$

A negative value means that high-regional-mean fields vary less across space. Because the bandwidths are equally spaced on the log scale, the arithmetic mean gives equal weight to every bandwidth in the prespecified five-scale grid. The primary summary averages record-specific ratios. Its asymmetry and sensitivity to small middle-day means motivate two further comparisons. The log-ratio analysis replaces R_{ymh} by $\log(\overline{Q}^{E}_{ymh} / \overline{Q}^{M}_{ymh})$ at the same three stages of averaging and back-transforms the final mean for interpretation. A second analysis uses the bounded symmetric contrast $2(\overline{Q}^{E}_{ymh} - \overline{Q}^{M}_{ymh}) / (\overline{Q}^{E}_{ymh} + \overline{Q}^{M}_{ymh})$. The three contrast measures are applied to identical fields and labels in the denominator stress test. We treat the log ratio as the principal robustness analysis and the bounded symmetric form as a secondary check.

2.2. Additive decomposition

Graph dispersion accumulates over edges. We obtain an exact and symmetric site allocation by splitting every undirected edge equally between its endpoints. For location i, define

$$q_{ih}(\mathbf{y}) = \frac{1}{4S_h} \sum_{j \neq i} w_{ij,h} (y_i - y_j)^2. \tag{7}$$

The identity is exact because the double sum counts every edge twice:

$$\sum_{i=1}^{n} q_{ih}(\mathbf{y}) = \frac{1}{2S_h} \sum_{i<j} w_{ij,h}(y_i - y_j)^2 = Q_h(\mathbf{y}). \tag{8}$$

The allocation is translation invariant and nonnegative for a single field. It also respects the graph geometry: a site receives contributions only from its incident weighted contrasts, while distant pairs carry little weight in a local graph. Equal edge splitting avoids assigning an arbitrary direction to an undirected spatial relationship.

Let $\overline{q}^{E}_{iymh}$ and $\overline{q}^{M}_{iymh}$ denote the high- and middle-day means within record (y, m), and let $\overline{Q}^{M}_{ymh} = \sum_i \overline{q}^{M}_{iymh}$. The site contribution to the record-level relative contrast is

$$c_{iymh} = \frac{\overline{q}^{E}_{iymh} - \overline{q}^{M}_{iymh}}{\overline{Q}^{M}_{ymh}}, \qquad \sum_i c_{iymh} = R_{ymh}. \tag{9}$$

The regime-change quantity c_{iymh} may be negative. Its denominator is the record-specific middle-day dispersion. Sites in months with different baseline variability therefore enter the final allocation on the same relative scale as the corresponding record-level estimator.

Let $\mathcal{Y}$ be the set of summers. With $H = 5$, the reported node allocation is

$$c_i = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \frac{1}{|\mathcal{J}|} \sum_{m \in \mathcal{J}} \frac{1}{H} \sum_{k=1}^{H} c_{iymh_k}, \qquad \sum_{i=1}^{n} c_i = \widehat{\theta}_{\mathcal{Y}}. \tag{10}$$

The order and weights in Equation (10) match the estimator: equal months within a summer, equal graph scales, and equal summers. Changing the spatial support changes the incident edges and hence the allocation. Accordingly, every main-text map is calculated from the 121-node graph. For a scale-specific map, retaining one h in Equation (10) yields the corresponding scale-specific contrast through the observed-node sum.

For display, we fit a low-rank thin-plate REML surface to each set of 121 observed-node values and evaluate it on a fine land-masked raster. Graph construction, estimation, uncertainty calculations and node-sum identities all use the observed nodes, which are overlaid on each map. A negative c_i allocates part of the regional contraction to edges incident to site i; the allocation measures that site's share of the pairwise contrast. The maps also show regime means of $y_{it} - \bar{y}_t$ and their difference. These spatially centred fields separate a change in spatial pattern from a shift in the daily regional mean.

To quantify the dominant north–south structure, let $\boldsymbol{\ell}$ be centred latitude and $\mathbf{z}_t = \mathbf{y}_t - \bar{y}_t\mathbf{1}$. Define

$$\widehat{\beta}_{th} = \frac{\boldsymbol{\ell}^\top \mathbf{L}_h \mathbf{z}_t}{\boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}, \qquad \mathbf{e}_{th} = \mathbf{z}_t - \widehat{\beta}_{th}\boldsymbol{\ell}.$$

Orthogonality in the $\mathbf{L}_h$ inner product gives

$$Q_h(\mathbf{y}_t) = \frac{\widehat{\beta}^2_{th}\boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}{2S_h} + \frac{\mathbf{e}_{th}^\top \mathbf{L}_h \mathbf{e}_{th}}{2S_h}. \tag{11}$$

High-minus-middle means of the two terms, divided by middle-day total dispersion, add exactly to R_{ymh}. More generally, for a centred spatial basis matrix $\mathbf{B}$, the Laplacian projection has coefficients $\widehat{\boldsymbol{\beta}}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^- \mathbf{B}^\top \mathbf{L}_h \mathbf{z}_t$, where $(\cdot)^-$ is a generalised inverse. Centred latitude is the reference structural summary. Further projections add longitude and then elevation, obtained by dividing the official invariant ERA5-Land geopotential field by standard gravity. We compare the total structured and residual energies of these nested column spaces.

A second decomposition separates the persistent monthly field from daily departures. Write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$, where $\boldsymbol{\mu}_m$ is the site-specific 35-summer mean field for month m. Then

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \tag{12}$$

The first term is constant within a month and cancels from the high-minus-middle numerator. The other two terms quantify change in anomaly energy and change in climatology–anomaly alignment. Dividing both differences by the middle-day total dispersion yields two components that add exactly to R_{ymh}. The signed cross term measures alignment between the monthly climatology and daily anomalies; a negative value indicates opposing spatial patterns.

3. Statistical inference

3.1. Randomisation by cyclic shifts

Days within a month are serially dependent, so each month-year record is randomised as a unit. Its complete daily outcome profile is rotated by a randomly chosen offset while the mean-WBT series and labels remain in place. This joint rotation preserves dependence between bandwidths, and the statistic is recomputed after every shift. The one-sided Monte Carlo p-value is

$$p = \frac{1 + \sum_{b=1}^{B} I(T_b^{\text{shift}} \leq T_{\text{obs}})}{B+1}. \tag{13}$$

Validity requires cyclic invariance, a stronger property than generic stationarity. For record r, let $X_r = (\bar{y}_{rt})_{t=1}^{n_r}$ denote the mean series and let D_r collect the H graph-dispersion values for each day, with $X = (X_1, \ldots, X_R)$ and $D = (D_1, \ldots, D_R)$. The operator C_{n_r} rotates all H columns of D_r together, and $G = \prod_r C_{n_r}$ is the group of all joint rotations. The null hypothesis is group invariance:

$$(D \mid X) \stackrel{d}{=} (gD \mid X) \quad \text{for every } g \in G. \tag{14}$$

Conditional on the observed regional-mean series, independently changing the circular phase of the complete multiscale dispersion profile in any month-year record must leave the joint distribution of all profiles unchanged. Under this null, the group-rank argument gives a finite-sample-valid lower-tail test; the proof and the plus-one Monte Carlo result are in Supplementary Section S1 (Phipson and Smyth, 2010).

Truncated time-shift tests replace wrap-around invariance with strict stationarity (Harris, 2020; Yuan and Shou, 2024), but their conservative finite-sample form requires at least 19 shifts on each side for a 0.05 cutoff. Records of 30 or 31 days leave no usable aligned window under that requirement, so we use cyclic shifts under Equation (14).

The product condition is stronger than marginal cyclic invariance of each record because it permits the records to be rotated independently. It specifies the conditional distribution of the full profiles, beyond their mean contrast. Common circulation or soil-moisture states across adjacent months, seasonal progression and large-scale climate drivers can violate independent phase invariance. A targeted simulation in Supplementary Section S11.4 studies one shared cross-record process and its effect on test size. Scale-wise weak family-wise error control and the development-record summaries are also detailed in Supplementary Section S1.

3.2. Evaluation design

We used the summers of 2015 and 2022 to inspect WBT fields and graph profiles and select the daily fields, event labels, spatial support, bandwidths and aggregation. The remaining 33 summers form the evaluation record $\mathcal{Y}_H$ in Equations (5)– (6). The available records identify the two development summers but leave their original acquisition rationale undocumented.

For the global cyclic-shift analysis, set $T_{\text{obs}} = \widehat{\theta}_{\mathcal{Y}_H}$. Each Monte Carlo transformation rotates the five-column dispersion profile within each of the 99 evaluation month-year records, then recomputes every record ratio and the equal-month, equal-scale and equal-summer mean, with $B = 99{,}999$.

The exploratory global cyclic-shift analysis followed estimation of the 33-summer contrast. The statistic is the evaluation-record target in Equation (6). The test uses the conditional product-invariance null in Equation (14).

Contrast magnitude is reported through the record mean, summer-specific contrasts, negative-summer count and leave-one-summer-out range.

Version 2 of the analysis protocol is dated 2 August 2026. It specifies the Student, lag-2 Newey–West and sign-test components and their joint decision rule. Supplementary Table S1 gives the dates, statistical assumptions and results of the primary and subsequent analyses.

3.3. Inference under temporal dependence

For comparison, a process-level reference treats the summer sequence as a stationary weakly dependent process. Under standard strong-mixing, moment and long-run variance conditions, its mean admits a central-limit approximation; a growing-bandwidth HAC estimator needs a stronger uniform autocovariance condition (Ibragimov and Linnik, 1971; Newey and West, 1987; Andrews, 1991). The full statement, ratio negative-moment conditions and proof are in the Supplement. At 33 summers, simulation shows that Student and HAC intervals can have substantial undercoverage under serial dependence. Both intervals are reported alongside the finite-record randomisation result.

4. Simulation study

The experiments assess whether the scale profile distinguishes different forms of spatial contraction and whether the associated inference remains calibrated. They contrast changes in anomaly amplitude, spatial correlation range and geographic gradient, using the observation geometry of the application. Further designs couple event intensity to the spatial field or introduce temporal dependence. Complete specifications and results appear in Supplementary Section S11.

4.1. Sensitivity to spatial structure

Sensitivity depends on the spatial form of the change. A reduction in anomaly amplitude affects dispersion throughout the bandwidth profile; range expansion acts predominantly at short distances, whereas gradient attenuation is concentrated at broad scales. These distinct signatures explain why an aggregate variance reduction alone provides an incomplete description of spatial change.

The comparator results follow this distinction (Figure 1). Neighbouring-pair semivariance is particularly sensitive to range changes, while ordinary variance is effective for gradient attenuation. Standardised autocorrelation measures lose sensitivity to pure amplitude changes because their normalisation removes that component. The graph profile maintains sensitivity across all three forms of contraction. Its advantage for the application is the joint interpretation of the scale signature and the spatial decomposition within a common dispersion measure. Binned variograms provide a closely related representation of distance structure and form the principal comparator in the empirical assessment.

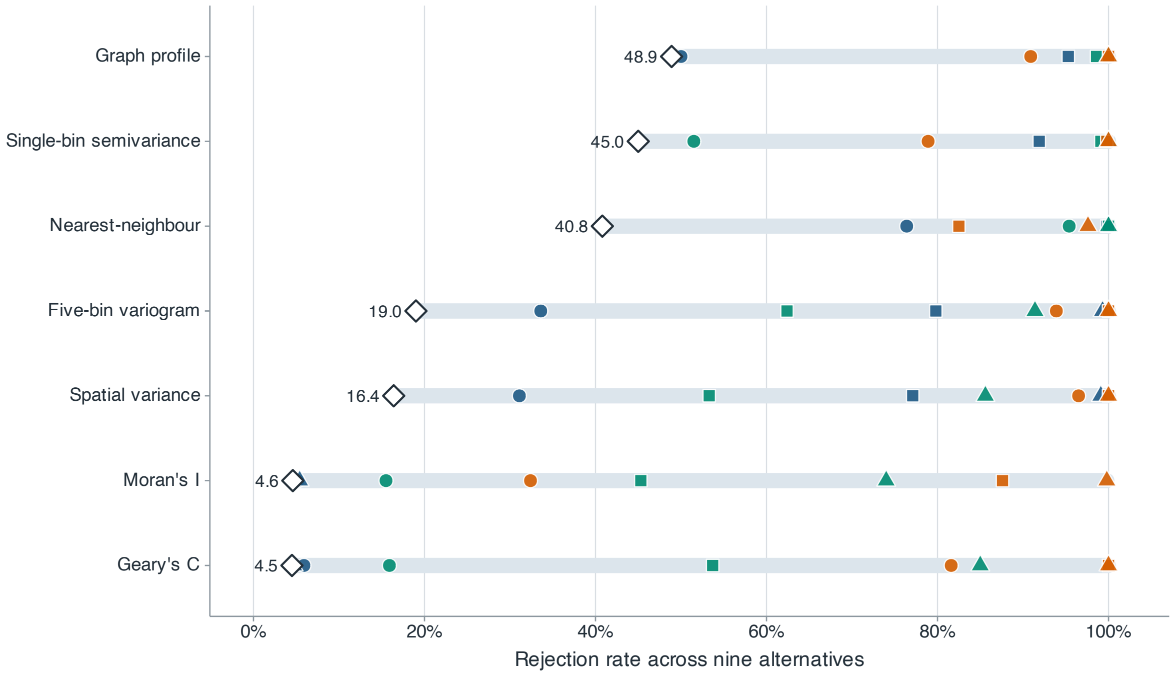

Figure 1 Power across nine simulated spatial alternatives. Blue, green and orange marks denote amplitude reduction, correlation-range expansion and gradient suppression; circles, squares and triangles denote weak, medium and strong changes. For each method, the pale segment spans power across the nine alternatives and the diamond marks the minimum.

Alt text: A range-and-dot plot compares seven methods. Each method has nine coloured and shaped points, a pale segment spanning their range and a diamond at the minimum; the ranges overlap substantially.

4.2. Calibration under temporal dependence

The randomisation procedure remains approximately calibrated in the simulated cyclic-invariant settings. Designs with shared structure in the regional mean and spatial field give similar calibration, including seasonal progression and selection of a daily regional peak. These results support use of the complete scale profile as the randomised object under the conditional invariance assumption. Their calibration depends on the joint structure of event labels and dispersion profiles, beyond the mean contrast alone.

Dependence between summers primarily affects uncertainty estimation. At the observed record length, the Student and HAC intervals under-cover when the summer sequence is positively correlated, while the mean contrast remains approximately centred. Preserving the actual evaluation calendar gives the same distinction. Thus removing the development summers has little effect on centring in these designs; estimating variation across a short dependent record is the main difficulty. The empirical interpretation accordingly combines the conditional randomisation result with the scale profile and the recurrence of the contrast across summers.

4.3. Stability of relative contrasts

Calibration of a randomisation test can coexist with imprecise effect estimation. In the heavy-tail designs, the raw dispersion ratio has much greater sampling variability than its log and bounded transformations, although their rejection behaviour is similar. The transformations reduce the influence of extreme daily energies on the relative comparison. This motivates reporting the raw contrast with transformation checks, so that the scientific interpretation rests on the common spatial pattern rather than on a single ratio scale (Supplementary Section S10).

5. Humid heat in eastern China

5.1. Study data

We analyse synchronous ERA5-Land wet-bulb temperature fields over 105–125°E and 20–42°N during June–August 1991–2025 (Muñoz-Sabater et al., 2021). The domain spans the pronounced subtropical-to-temperate latitudinal gradient while excluding most of the Tibetan Plateau. The primary lattice contains 121 land sites, with a nested 465-site lattice for spatial-resolution and coverage comparisons (Figure 2). The evaluation sample excludes the two summers used for method development, as described in Section 3.

Wet-bulb temperature is derived from air temperature, dew point and surface pressure using the thermodynamic formulation of Bolton (1980) and Davies-Jones (2008). Each daily field contains simultaneous values at the hour of the maximum regional mean. This common observation time gives the coverage measures their interpretation as concurrent spatial extent. Within-month high and middle groups follow the definitions in Section 2. Supplementary Section S14 records the conversion and field-selection details.

The 1950–1990 ERA5-Land record provides a temporal extension at the same sites. NOAA observations provide a measurement comparison on a sparse station network, using matched event times and spatial support (Smith et al., 2011). Their distinct temporal and spatial supports are retained in the interpretation of the results.

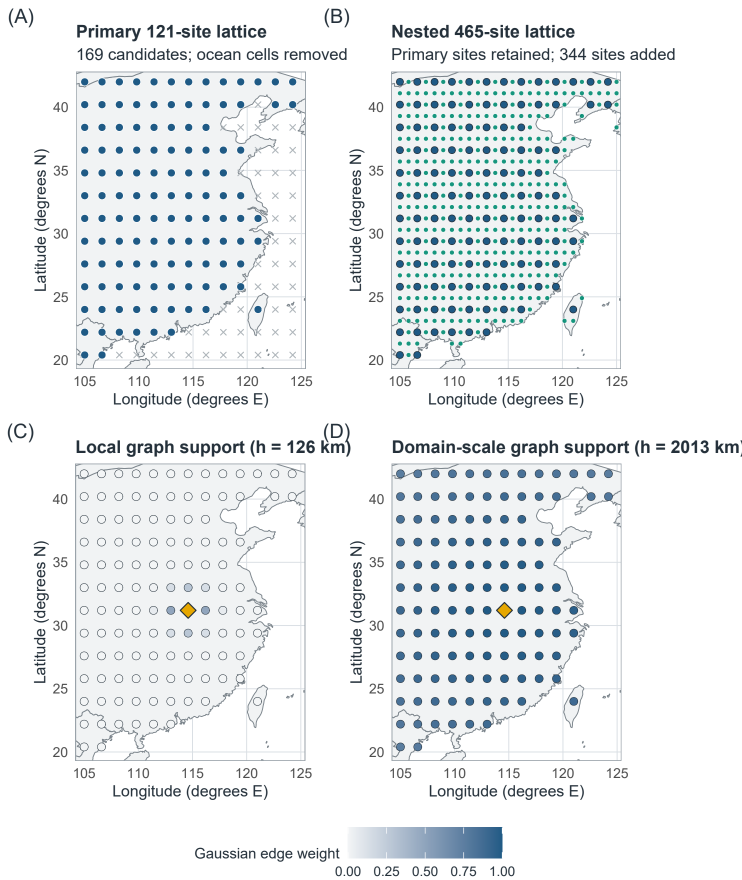

Figure 2 Spatial sampling and graph support. (A) Analysis lattice with 121 land sites from 169 candidates. (B) Nested lattice adding 344 land sites while retaining every analysis site. (C,D) Gaussian edge weights from one reference site under the 126- and 2,013-km graphs, showing how bandwidth changes the geographic support at fixed coordinates. Basemap: Natural Earth 1:50m, accessed through the R `maps` package.

Alt text: Four eastern-China maps compare the analysis and nested site lattices, then show weights concentrated near one reference site under the 126-km graph and spread across the domain under the 2,013-km graph.

5.2. Spatial extent of humid heat

We quantify simultaneous coverage on the nested grid at the original daily peak times. Let a_i denote the land area represented by site i and $a_+ = \sum_i a_i$. Total exceedance area and the largest connected exceedance region, expressed as fractions of represented land, are

$$A_t(u) = \frac{\sum_i a_i I(y_{it} > u)}{a_+}, \qquad C_t(u) = \frac{\max_{B \in \mathcal{B}_t(u)} \sum_{i \in B} a_i}{a_+}, \tag{15}$$

where $\mathcal{B}_t(u)$ comprises four-neighbour connected components and $C_t(u) = 0$ for an empty exceedance set. The represented area is 3.221 million km^2; multiplying either fraction by this area gives its geographic extent. These measures describe concurrent humid heat at the sampling resolution, following the emphasis on event extent in Healy et al. (2025) and Skinner et al. (2025).

The analysis retains the complete temperature-threshold curve fixed from the development summers. High-minus-middle comparisons are averaged equally over months and evaluation summers, with descriptive uncertainty obtained from blocks of annual contrasts. Supplementary Section S13 gives the land-area, threshold and connectivity definitions.

To assess the information retained by spatial summaries, we compare a regional-mean and seasonal baseline with a stronger model containing spatial variance, the signed latitude gradient and the direct primary-site exceedance fraction. Graph and binned-variogram additions describe scale structure relative to total variance. The regression specification and blocked temporal validation are common to these comparisons (Roberts et al., 2017); interpolation provides a reference using the full primary field. The additional-site area comparison assesses coverage beyond the original sample locations. These comparisons isolate the incremental value of distance-resolved summaries for the same coverage target.

5.3. Scale dependence of spatial contraction

High regional wet-bulb temperature is associated with attenuation of broad geographic contrasts. The average relative dispersion contrast is -7.28%, but the more informative feature is its increasing magnitude with bandwidth (Figure 3; Table 2). The profile indicates that the decrease also detected by ordinary spatial variance is concentrated in broad geographic structure, with a smaller local component. This scale dependence provides the basis for the structural interpretation below.

The contrast recurs across evaluation summers and is supported by the protocol comparisons and the subsequent exploratory cyclic-shift analysis (Supplementary Sections S1 and S3). The latter evaluates the conditional invariance null in Equation (14). The principal result concerns differences between thermal states within summer; the estimated change in this contrast over calendar years remains uncertain.

The earlier climate record reproduces the concentration of contraction at broad scales (Supplementary Section S7). Its consistency with the evaluation period supports a recurrent spatial feature of high-mean humid heat. The extension shares the reanalysis system, including the preliminary forcing used for its earliest segment, so its value lies in the temporal persistence of the scale pattern.

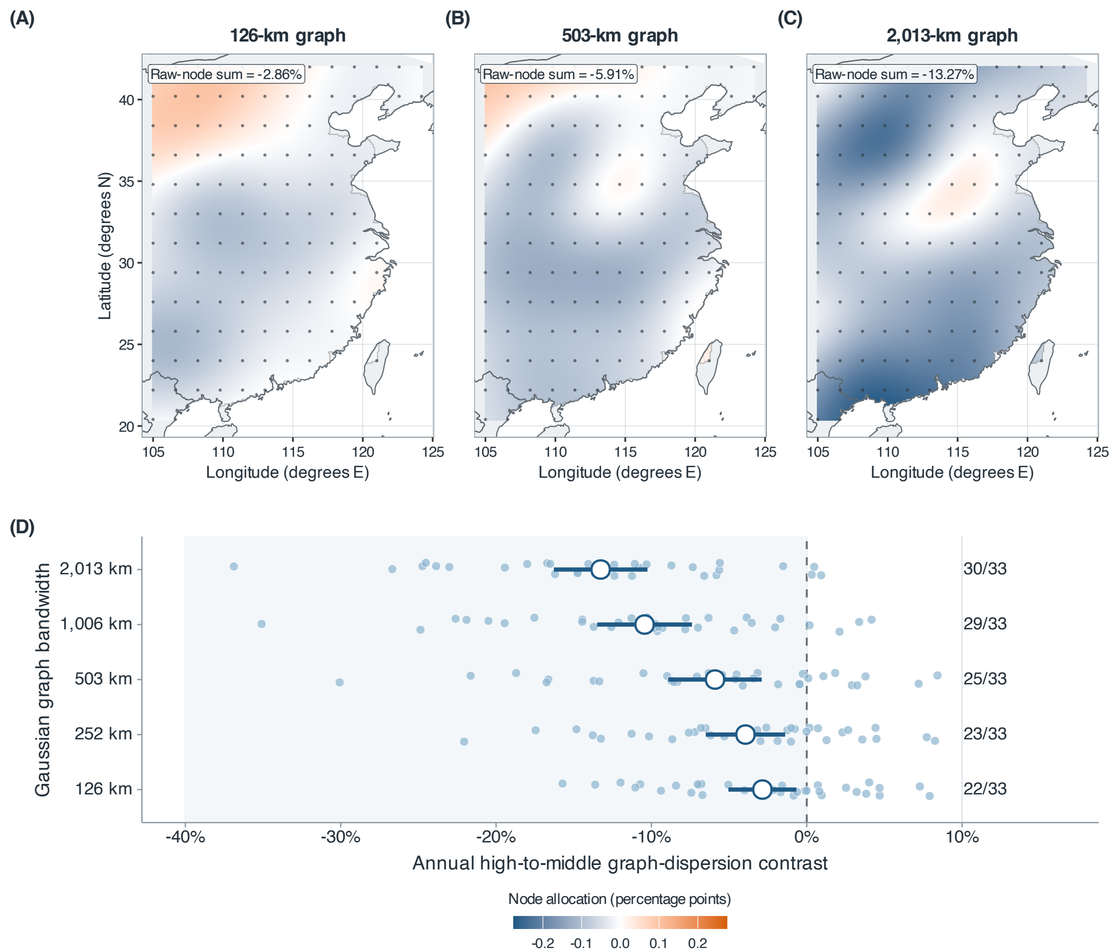

Figure 3 Scale-specific node allocations and summer contrasts for the 33 evaluation summers. (A–C) Allocations for the 126-, 503- and 2,013-km graphs share a symmetric colour scale; their observed-node sums are -2.86%, -5.91% and -13.27%. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes. (D) Small circles show summer-specific contrasts at all five bandwidths; large circles and bars show means and 95% descriptive (t)-reference intervals based on between-summer variation. Labels give the numbers of negative summers.

Alt text: Three smoothed eastern-China maps, each overlaid with the 121 analysis sites, show increasingly coherent negative node allocations at 126, 503 and 2,013 km. A lower panel shows the 33 annual contrasts at all five bandwidths, with mean contrasts and negative-summer counts increasing in magnitude with bandwidth.

5.4. Geographic extent of simultaneous humid heat

The attenuation of broad spatial contrasts accompanies a larger connected humid-heat footprint (Figure 4). At 24°C, the high-minus-middle differences in total and largest connected area are 12.91 and 12.57 percentage points, respectively, each corresponding to approximately 400,000 km^2 of represented land. The coverage contrast therefore includes substantial differences in the extent of contiguous humid-heat regions.

Threshold dependence is central to this interpretation. The area separation is greatest around 23–24°C and narrows towards the highest thresholds. The geographic expression of high regional humid heat consequently depends on the temperature level being considered. Spatial contraction alone does not determine this curve: regional mean and spatial pattern change together between the groups. Direct coverage measurement establishes the extent of their combined association.

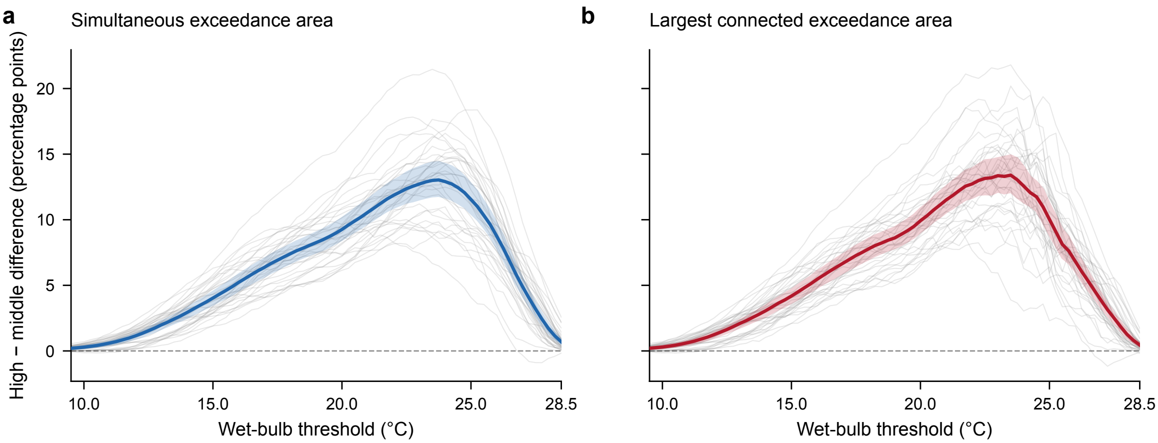

Figure 4 High-minus-middle differences in simultaneous humid-heat coverage at the original daily peak times. (a) Total exceedance area. (b) Largest four-neighbour connected exceedance area. Thick curves average months and 33 evaluation summers equally; thin grey curves show every summer. Shading gives pointwise 95% percentile intervals from 1,999 resamples of five-year calendar blocks. All 77 development-defined temperature thresholds are shown. Areas refer to the 3.221-million-km^2 represented land support.

Alt text: Both area-difference curves rise from small positive values at low thresholds to approximately thirteen percentage points near 24 degrees Celsius, then decline toward the upper end of the threshold range. Grey annual curves show variation between summers, including some negative values at high thresholds.

The comparison of spatial summaries identifies where most coverage information resides (Table 1). The principal reduction in error occurs when the regional mean is supplemented by simple spatial summaries. Distance-resolved predictors provide small further improvements, with graph and variogram representations performing similarly. The same conclusion holds for the additional-site reference. Thus coverage estimation and structural interpretation have different informational requirements: the simple summaries describe geographic extent well, while the graph profile connects regional dispersion to its spatial scales and components. The magnitude and annual consistency of the incremental gains are documented in Supplementary Section S13.

5.5. Latitudinal structure of the contraction

The broad-scale association is concentrated in the persistent latitudinal gradient. Relative to each day's regional mean, high-day fields are warmer in the north and cooler in the south than middle-day fields (Figure 5). The spatial allocations therefore identify attenuation of a familiar climatic contrast as the main geographic expression of the regional statistic.

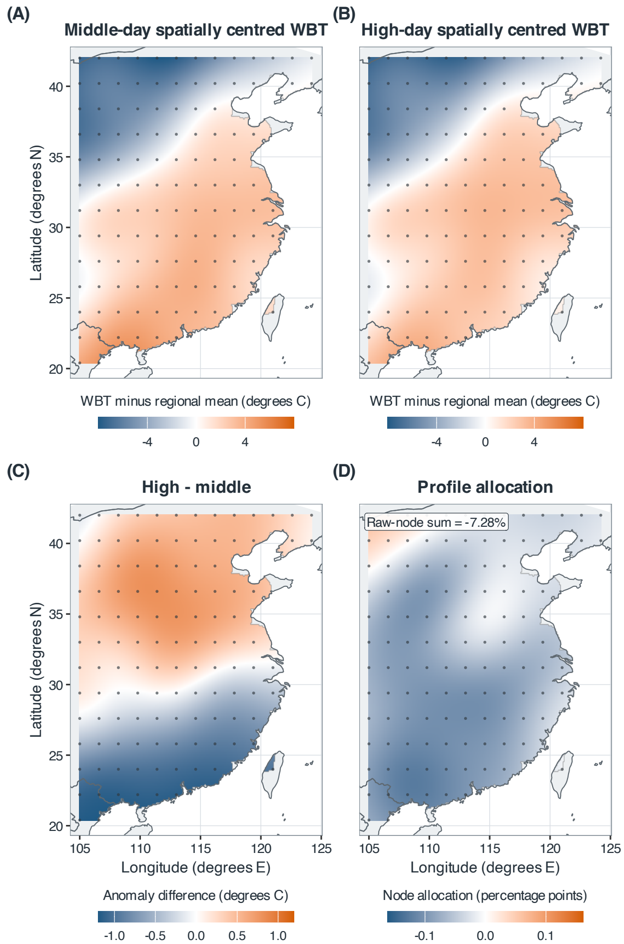

Figure 5 Spatial structure of the 121-site evaluation contrast. (A,B) Middle- and high-day spatially centred fields; (C) their difference; (D) exact observed-node allocations averaged over the five graph scales. The 121 allocations sum to -7.28%, identical to the five-scale estimand. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes.

Alt text: Four smoothed eastern-China maps derived from 121 analysis sites, shown as dots. The spatially centred surfaces show a weaker north–south contrast on high days; the high-minus-middle surface changes from positive in the north to negative in the south, and most observed-node allocations are negative, with the strongest negative values concentrated in the central and southern parts of the domain.

The climatology decomposition gives this pattern a more specific interpretation. High-day anomalies oppose the monthly climatological pattern more strongly, while their own broad-scale energy changes little (Figure 6). The decline in total dispersion appears primarily in the signed alignment between these two fields. Regional coherence can therefore increase through compensation of a persistent gradient, with comparatively little change in the magnitude of transient spatial anomalies.

The projection analyses support this interpretation. Latitude accounts for most of the resolved broad geographic component; including longitude and elevation preserves the conclusion about the total structured component. Re-estimating climatology without the evaluated summer has little effect on the decomposition. Together, these checks associate the contraction with the observed climatic gradient rather than a particular fitted reference field (Supplementary Section S5).

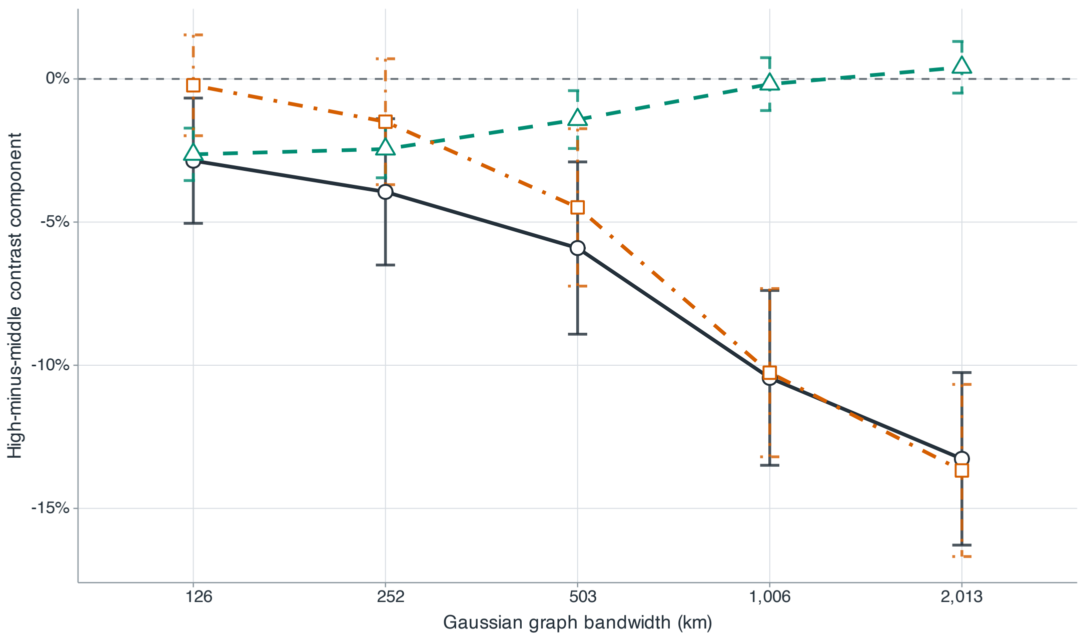

Figure 6 Exact decomposition of the raw-field graph contrast. Dark circles, teal triangles and orange squares denote total change, anomaly-energy change and the climatology–anomaly cross component; bars are 95% descriptive intervals. Monthly-climatology energy cancels within each record.

Alt text: Across five bandwidths, dark circles become increasingly negative. Teal triangles remain near zero at broad scales, while orange squares closely follow the negative total.

5.6. Sensitivity of the structural interpretation

Graph weighting, distance definitions and sampling density change the estimated magnitude while preserving the broad scale ordering. Domain changes are more consequential because they alter the geographic gradient represented by the field. The contrast is thus a property of the specified spatial support, and its scale profile is more stable across these choices than any single regional percentage (Figure 7; Supplementary Section S6).

Field definition has a clearer substantive effect. Removing site climatology substantially attenuates the contrast, consistent with the structural decomposition. Fixed-hour and daily-mean fields retain the negative association, and the continuous-intensity analysis extends it beyond the quartile grouping. Calendar matching gives a weaker contrast on its restricted set of eligible day pairs while preserving scale ordering. These comparisons distinguish variation in the climatic field from sensitivity to how days are selected; the underlying definitions and results are given in Supplementary Sections S9, S12 and S14.

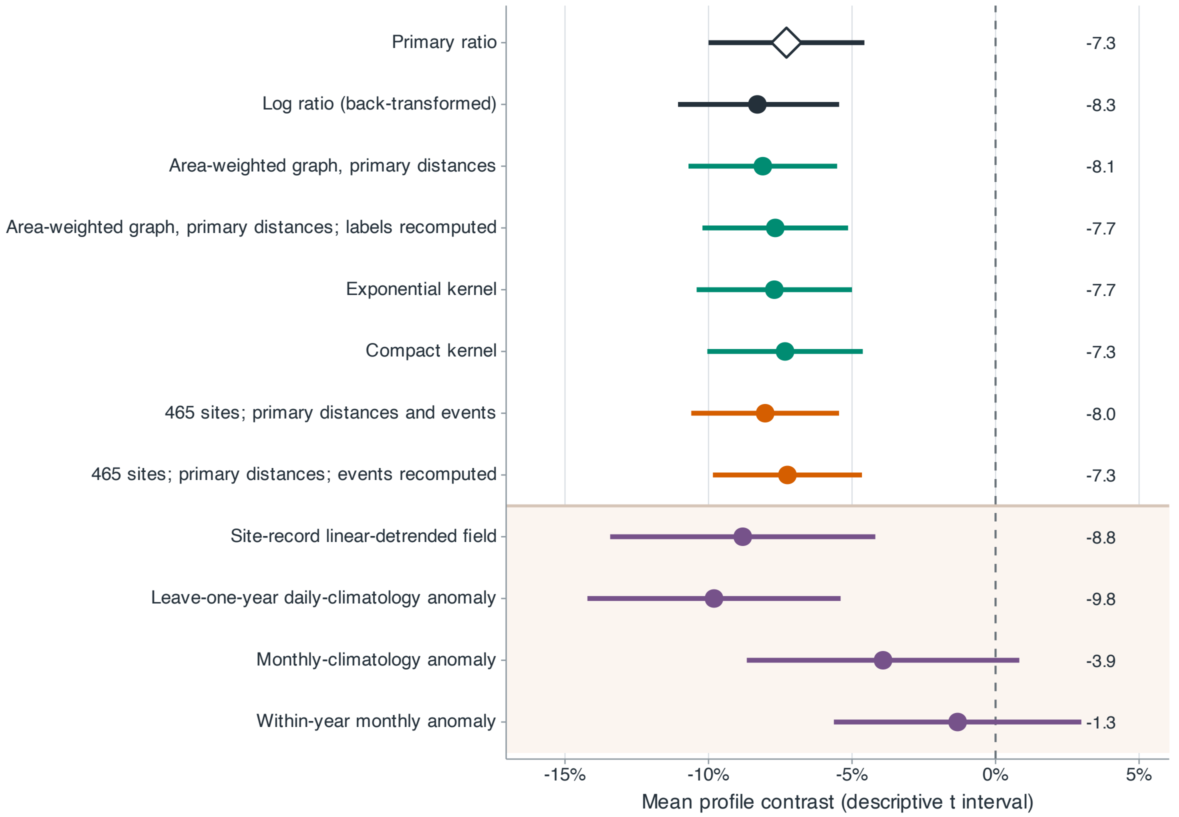

Figure 7 Robustness of the mean profile contrast. Dark rows denote the main and transformation analyses; green, orange and purple rows denote graph construction, spatial resolution and field definition. Points and bars are estimates and descriptive t-reference intervals based on between-summer variation.

Alt text: A forest plot shows mostly overlapping negative estimates across transformation, graph and resolution choices. Two purple-shaded anomaly-field rows lie closest to zero.

The station comparison supports the broad geographic interpretation (Figure 8). Reanalysis and station observations show concordant contraction at the scales supported by the station network, whereas local contrasts are less precisely determined. The broad direction also persists with station-defined events and fixed spatial support. Stricter availability requirements widen uncertainty by reducing the retained record. The comparison therefore provides complementary measurement evidence for the broad pattern at common events, with the greatest uncertainty at short distances (Supplementary Section S8).

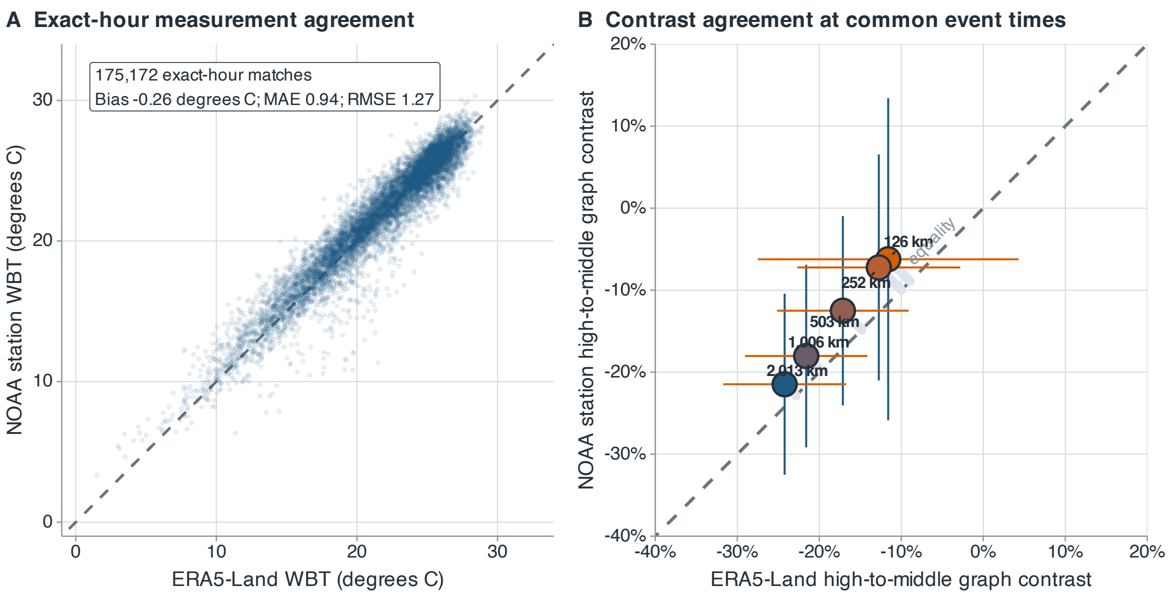

Figure 8 NOAA station and ERA5-Land agreement in ten station-comparison summers. (A) Exact-hour WBT at 28 stations, with summaries from all 175,172 matches. (B) High–middle graph contrasts at the five bandwidths, using ERA5-Land peak times and labels and identical station subsets. Dashed diagonals denote equality. Horizontal and vertical bars in panel B are paired descriptive t-reference intervals based on between-summer variation.

Alt text: Two panels compare NOAA stations with ERA5-Land. Exact-hour WBT points in the left panel cluster around the equality line. In the right panel, station and ERA5-Land high–middle contrasts are negative at all five bandwidths and become more negative as the bandwidth increases; station estimates are slightly closer to zero.

6. Discussion

The dominant spatial change on high regional wet-bulb temperature days is an attenuation of broad climatic differences. At these scales, the daily anomaly field retains comparable energy, but its alignment with the monthly climatology shifts towards compensation of the latitudinal gradient. This distinction gives a climatic interpretation to the dispersion result: a smaller regional variance can reflect weaker persistent geographic contrasts even when transient spatial variability remains substantial.

The coverage analysis establishes the geographic extent associated with these thermal states. High-mean days have more extensive simultaneous exceedance and larger connected humid-heat regions, with the greatest area separation in an intermediate part of the temperature range. Regional mean, spatial dispersion and exceedance area describe different aspects of this state. Their joint interpretation distinguishes a broad humid-heat footprint from the scale and structure of variation within it. The observed coverage difference combines changes in regional level and spatial pattern.

The statistical contribution is an additive formulation that connects the regional contrast to its scale profile and spatial components. Site allocations and projections recover the same relative quantity used in the regional comparison, allowing the climatic gradient to be examined within that common measure. The application shows the interpretive value of this connection. Simple spatial summaries already retain most information about coverage, while the graph decomposition identifies where in scale and geographic structure the dispersion change occurs. Structural interpretation and coverage estimation are therefore distinct uses of the spatial data.

The evidence is strongest for broad spatial structure on the analysed support. Sampling resolution limits local interpretation, and dependence between summers affects the precision of inference from the available record. Long-term changes in China's latitudinal wet-bulb temperature pattern have been associated with heterogeneous water-vapour changes (Wang et al., 2024). Our event-scale comparison identifies a recurrent spatial configuration of high humid heat: weaker broad climatic contrasts together with a more extensive connected footprint.

Data availability

ERA5-Land is available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land`). The NOAA ISD station archive (Smith et al., 2011) is available at `https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database`. Code and retained results for this version are archived at `https://github.com/Stork343/eastern-china-wet-bulb-contraction/releases/tag/jrssc-submission-v7-2026-09-08`. The accompanying `jrssc_reproducibility_bundle.zip` contains the current manuscript, analysis code, protocols, site and station manifests, and retained results, including the September coverage comparison. Its `code/revision_202609/README.md` gives the application-analysis entry points; `output_revision_application/` contains the fixed specification, coverage curves, held-out predictions and annual comparisons. Provider-data download and field-construction programs are included. The raw reanalysis and station archives are obtained from the providers above.

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Conflict of interest

The authors declare no conflict of interest.

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Table 1. Held-out mean absolute error in area percentage points. Errors are averaged over all 77 thresholds (9.5–28.5°C at 0.25°C steps), then equally over months and 33 evaluation summers. All-day errors use 3,036 June–August daily fields in 1991–2025 excluding 2015 and 2022; high-day errors use the 792 original upper-quartile days. The last two columns use the 344 additional sites. All inputs come from the original 121 sites.

	Exceedance area		Largest component		Additional sites	
Method	All	High	All	High	All	High
Mean and season	2.794	2.291	3.070	2.593	2.818	2.314
Strong simple baseline	0.827	0.747	1.389	1.268	1.028	0.929
Simple + variogram	0.815	0.741	1.366	1.266	1.013	0.922
Simple + graph	0.820	0.743	1.379	1.265	1.020	0.925
Spatial interpolation	1.398	1.414	1.823	1.859	1.861	1.882

Table 2. Physical scale of the evaluation-period graph contrasts. h is the Gaussian bandwidth. $\bar{d}_w$ is the edge-weighted mean pair distance, and N_{eff} is the Kish effective edge count. Graph contrasts are relative changes in weighted squared differences. RMS contrasts average record-specific ratios of $(2Q)^{1/2}$; Negative is the number of negative summer graph contrasts.

h (km)	$\bar{d}_w$ (km)	N_{eff} (edges)	Graph contrast (%)	RMS contrast (%)	Negative (/33)
126	200	366	-2.86	-1.54	22
252	319	1,138	-3.94	-2.14	23
503	549	3,223	-5.91	-3.25	25
1,006	826	6,005	-10.44	-5.70	29
2,013	992	7,109	-13.27	-7.22	30